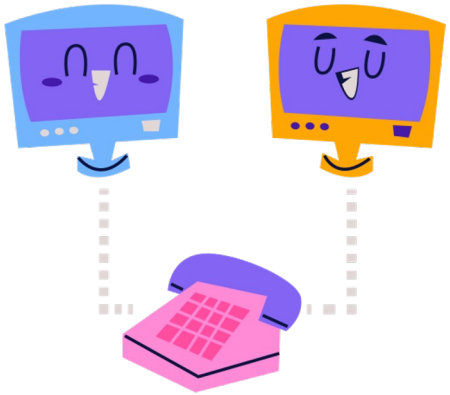


Introduction to Computer Networks

By Maimoona Khilji

Meet your Trainer

NAME

MAIMOONA KHILJI

EDUCATION

MS DATA SCIENCE (2025-27)
BS DATA SCIENCE (2019-23)
from the Institute of Management Sciences

EXPERIENCE

DATA SCIENTIST | Atomcamp
Visiting Faculty | Institute of Management Sciences

ACHIEVEMENTS

DISTINCTION HOLDER + MICROSOFT-CERTIFIED
Straight A+ student, speaker and panelist in various tech forums, recipient of multiple merit-based scholarships.



Introduce yourself



Your Good Name



Your background (city)



Your Favourite Subject



Your Hobbies or interests

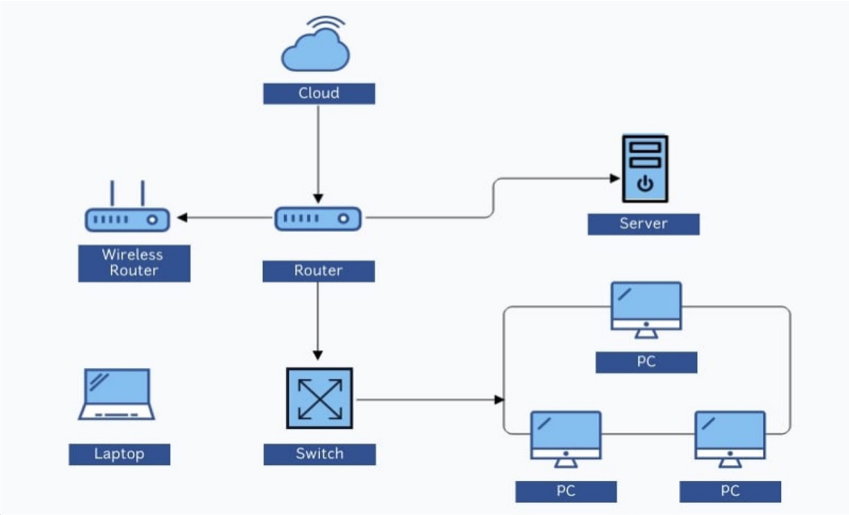
Learning Objectives

By the end of this week, students should be able to:

- Explain what a computer network is and why networks are important.
- Describe basic networking concepts and terminology.
- Understand wired/wireless networks.
- Differentiate between LAN, MAN, WAN, Internet, and Intranet.
- Describe the OSI Reference Model and TCP/IP Model at a basic conceptual level.

Computer Networks

A computer network is a collection of two or more connected devices that can communicate with each other and exchange data or share resources.



Why Do We Need Computer Networks?

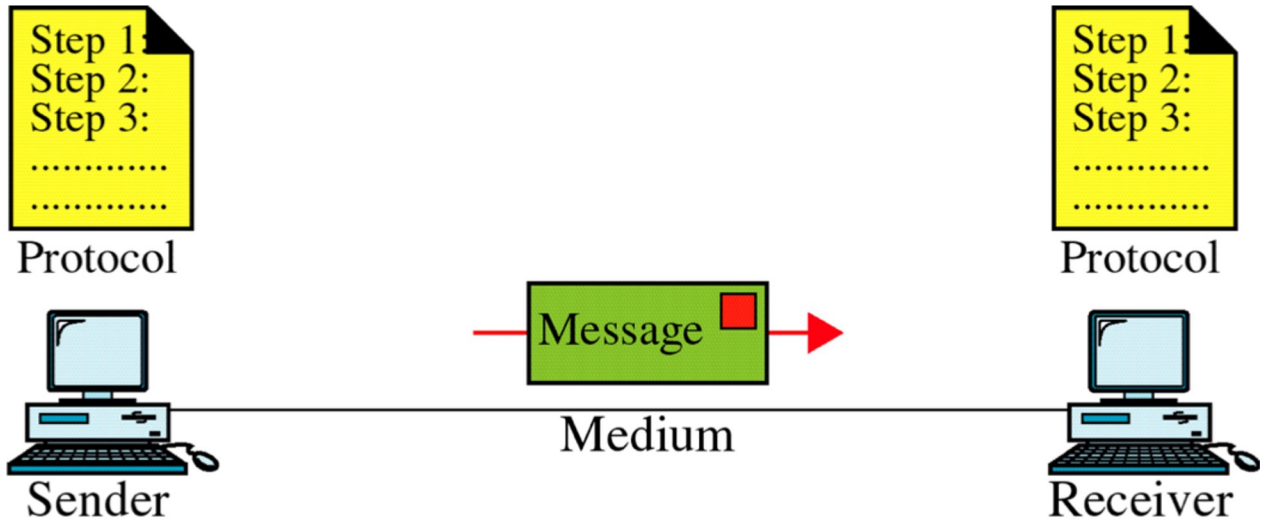
Networks are an essential part of modern computing because individual computers cannot efficiently operate in isolation.

1. Communication
2. Resource Sharing
3. Data Sharing
4. Centralized Services
5. Internet Access

How Does Network Communication Work?

A data communication system is made up of 5 components:

1. Message
2. Sender
3. Receiver
4. Medium
5. Protocol

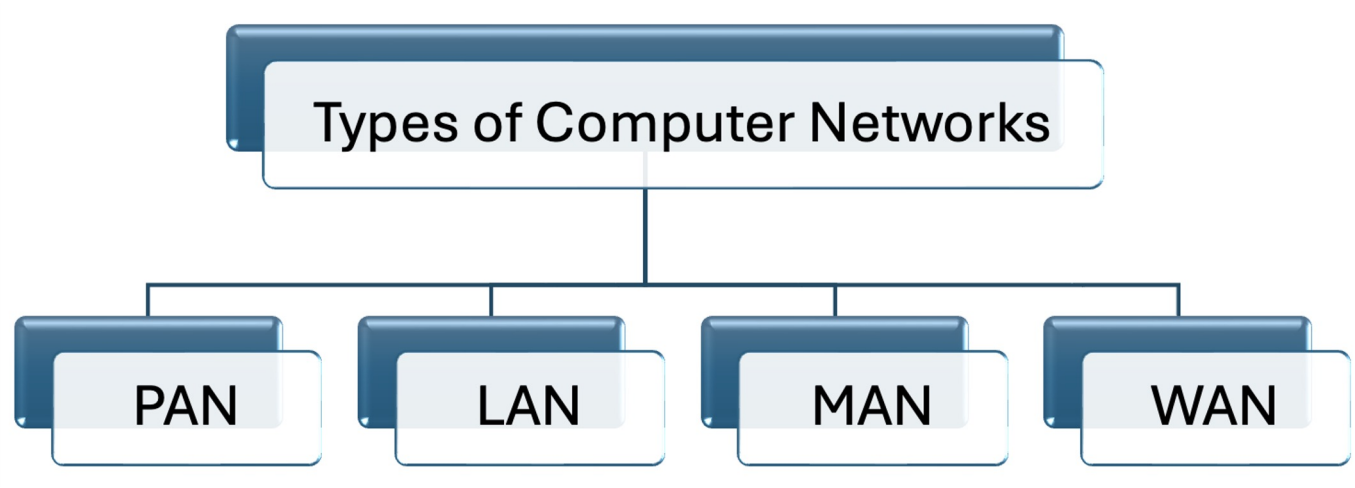


Wired and Wireless Networks

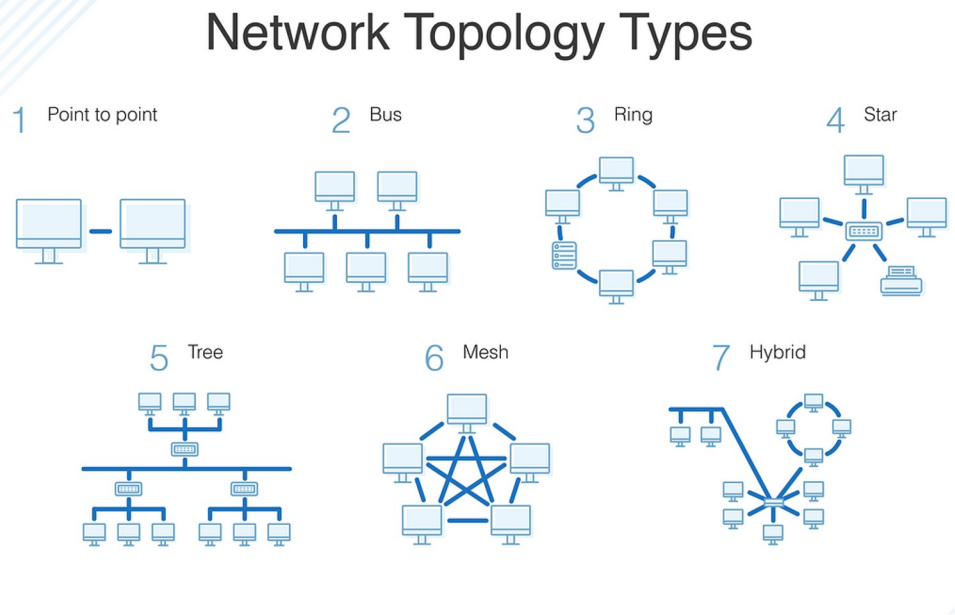
1. **Wired Network**
   It uses a physical cable to connect devices. Common technologies include Ethernet.
   Advantages
   - Stable connection
   - Generally reliable
   - Suitable for fixed devices

1. **Wireless Network**
   It allows devices to communicate without a physical network cable between the device and wireless access point. Wi-Fi is a common example.
   Advantages
   - Mobility
   - Easier connection for mobile devices
   - Convenient for users

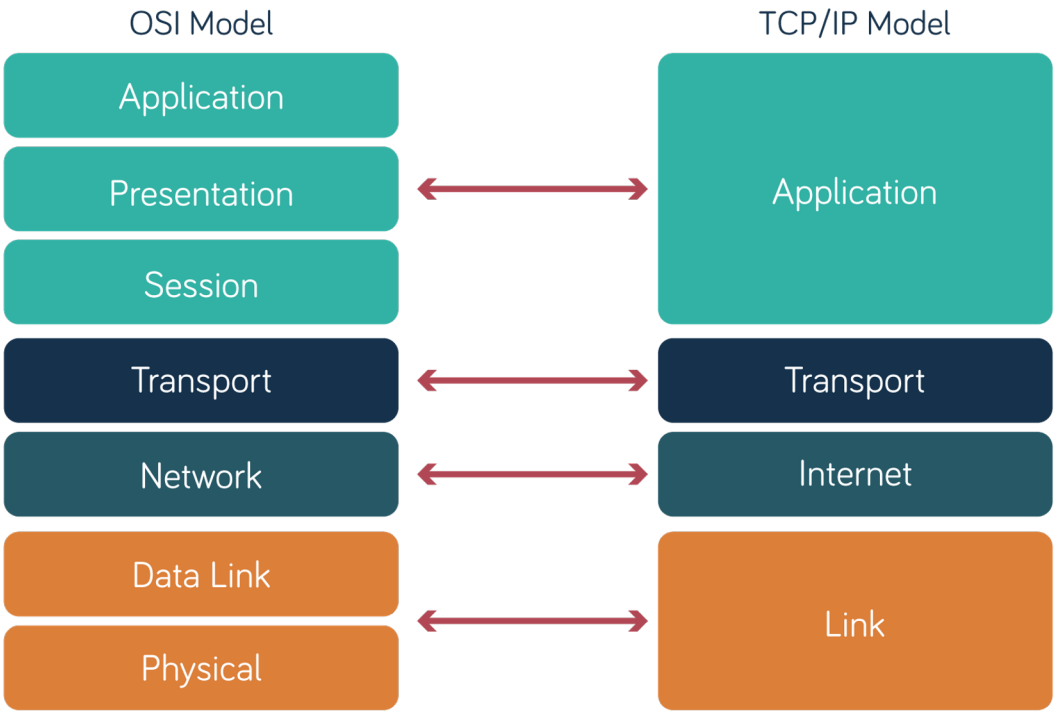
Network Types Based on Geographic Area



Network topology



OSI vs TCP/IP


OSI Model
TCP/IP Model
Application
Presentation
Session
Transport
Network
Data Link
Physical
Application
Transport
Internet
Link

Thanks!